

Patrick Pelegrino

FullStack Software Engineer

Iloilo, Philippines 5023
pelegrinompatrik@gmail.com
[linkedin.com/in/patrick-pelegrino](https://www.linkedin.com/in/patrick-pelegrino)
<https://github.com/madpurson>

EDUCATION

University of the Philippines Visayas

Bachelor of Science in Computer Science

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Operating Systems, Game Development, Mobile Development, Web Development

Central Visayas Institute Foundation

Science, Technology, Engineering, and Mathematics Strand

- **Honorific Scholar:** First Semester – Last Semester
- **Honors:** With High Honors

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, PHP, Dart, Swift, C++; Node.js, Express.js, React.js, X-86 Assembly

Software and OS: Android Studio, Visual Studio Code, Visual Studio, Jupyter Notebook, Git, Linux, Unity, Figma, Puppeteer, Scrapy, Docker, Postman, CI/CD (Vercel, Render), Redux, nodemailer, node-cron, Bruno, Jest, Maestro

Frameworks: React, React Native, Bootstrap, TailwindCSS, Flask, Flutter, Flask, Arduino

Database: MongoDB, SQLite, Firebase Firestore, MySQL, Oracle

PROFESSIONAL EXPERIENCE

Powerfan, Junior Frontend Developer | Seoul, Korea (Remote)

Aug, 2025 – Dec, 2025

- Optimized application performance by implementing Redux with optimistic rendering, significantly improving UI responsiveness and perceived performance.
- Engineered a comprehensive testing suite, writing unit and integration tests with Jest and end-to-end tests with Puppeteer, which reduced production bugs by over 30%.
- Established and maintained CI/CD pipelines using GitHub Actions, automating testing and deployment processes to enable faster, more reliable feature rollouts.
- Performed rigorous backend API testing and validation using Bruno, creating a collection of test requests to ensure endpoint reliability and data integrity before frontend integration.
- Configured and monitored cron jobs for automated daily testing cycles of both frontend and backend services, proactively identifying failures and improving system stability.
- Led frontend bug fixes and improved API handling logic, enhancing application performance and user experience.
- Identified and resolved critical code hazards including null safety issues in core components, enhancing application stability and preventing potential runtime errors.

Axiom, Backend Developer | Iloilo, Philippines

Dec, 2024 – May, 2025

- Developed backend infrastructure for a mobile app using Flask, managing data pipelines and logic for user sessions, recommendations, and content delivery.
- Integrated SQLite for local persistence and Firebase Authentication for secure login and identity management across multiple user roles.
- Built and deployed a RESTful API suite for data handling and communication between the app's client-side interface and backend services.
- Designed and implemented a content-based recommendation system using TF-IDF vectorization, enhanced with contextual outputs via LLaMA 2, improving user engagement and personalization.
- Coordinated closely with mobile developers to support rapid iteration and robust performance under concurrent load.

Powerfan, Frontend Developer & QA Intern | Seoul, Korea

Dec, 2024 – Feb, 2025

- Developed and optimized a React.js web application UI with Bootstrap, focusing on responsiveness, accessibility, and performance across devices.
- Integrated AI-powered backend APIs into the frontend interface, enabling interactive features and dynamic content delivery driven by generative models.
- Conducted automated testing using Puppeteer to validate feature stability, detect regressions, and ensure smooth deployments.
- Collaborated with cross-functional teams to align frontend behavior with backend API logic, improving feature rollout and QA workflows.
- Maintained test case documentation and contributed to improving frontend code quality and usability.

MECS Software Solutions, Backend Developer Intern | Iloilo, Philippines

June, 2024 – August, 2024

- Developed a multi-role Point-of-Sale (POS) system with modules for sales tracking, inventory, payments, and reporting dashboards.
- Used Drift ORM with SQLite to build a performant local database with optimized queries for real-time transactions.
- Authored RESTful API endpoints to connect frontend and backend, supporting dynamic updates for multiple

concurrent cashier logins.

- Participated in daily stand-ups and team reviews, contributing ideas to improve UI integration and user access flow.
- Delivered a working demo with documentation for potential deployment in small retail businesses.

PROJECT EXPERIENCE

Disclaimer: Due to legal and ownership constraints, professional work is excluded from this list. The projects below are personal and side projects

CHIRON – Health Misinformation Detection System, *Web Developer & Data Scientist | Special Problem Project*

Aug, 2024 – May 2025

- Built a full-stack misinformation detection system using DistilBERT, capable of classifying Philippine health news credibility with high recall and precision.
- Developed 5 modular backend APIs for article scraping (Puppeteer, BeautifulSoup), preprocessing (tokenization, TF-IDF), classification (DistilBERT, SVM, RF, Naive Bayes, LSTM), and response delivery.
- Integrated a React-based frontend with REST APIs to allow users to submit links and view results in real-time; deployed via Render and Vercel.
- Achieved 97–99% accuracy across traditional and deep learning models, with lightweight deployment optimized for latency and repeated queries.

MeowgAw – Pet Adoption & E-Commerce Platform, *Web Developer | Group Project*

Aug, 2023 – May, 2024

- Built a MERN-stack application with user-friendly UI and robust backend architecture.
- Managed pet listings, user profiles, and transactions to support local adoptions.

Dice-cider – DND-Inspired Decision-Making App, *Mobile Developer | Group Project*

March 2024 – May 2024

- Built a Flutter-based mobile app simulating a DND-style dice to help users make light-hearted, randomized decisions.
- Integrated Firebase and Cloud Firestore for storing user prompts and logging outcomes in real time.
- Designed a playful and intuitive UI, emphasizing accessibility and responsiveness across Android devices.

Katipunan – Digital Repository of Precolonial Filipino Culture, *Web Developer | Group Project*

Feb, 2023 – May 2023

- Built and deployed an online platform that archives literature, art, and knowledge from the precolonial Philippines, inspired by the spirit of “tipun” (gathering).
- Engineered end-to-end web functionality using PHP, JavaScript, MySQL, and CSS, ensuring fluid integration between interactive UI elements and robust backend logic.
- Designed responsive front-end interfaces and structured databases to optimize user experience and content discoverability across the site.

